

DECISION ANALYTICS

Analytical Case Study Report

Using Predictive Analytics to Improve Peak Electricity Demand Forecasting and Generation Planning at Ceylon Electricity Board, Sri Lanka

Ceylon Electricity Board Peak Demand Risk Case Study

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Abstract

How the Decision Analytics can Support the CEB (Ceylon Electricity board) in improvement of peak electricity demand that is short term and the generation planning. The problem that is selected has not been treated as a machine learning exercise that is generic but is developed by using the electricity evidence of Sri Lanka. Where the total generation, the average daily generation that is rising, peak demand that is high in night, and the rooftop solar export growth that is rapid are shown. The total generation of CEB was increased to 16802 GWh in the year 2024. And the per day average daily generation reaching 45.91 GWh, 2673 MW night time max demand and increasing of roof top solar exports to 867GW. A planning environment is created where the demand, the renewable variability, the availability of hydro and thermal generating together the decisions should be considered by these indicators.

Firstly, the domain of the organization, the context of the problem and the evidence before the aim is stated and the objectives. Then using keywords, the problem is broken down, the questions 5W1H, the data factors, the analysis of feasibility and between the discovery of knowledge, data mining, machine learning, data-science and making organizational decisions. A synthetic daily operational dataset is used in the analytical practical solution because of CEB SCADA that is confidential. No public availability of dispatch and smart meter datasets. Declared as synthetic and calibrated with official CEB 2024 anchor values for keeping the modelling demonstration connected to case study – Sri Lanka.

For classifying daily peak demand as a low risk, using weather medium or high, calendar, demand and generation variables a Random Boost classifier is developed. 68.48% test accuracy was reached by the model. Which depicts decision signal that is useful but is also confirmed that without the real data from CEB for operational deployment the prototype is not ready.

In this report it concludes that from a dashboard that is decision supporting that has the combination of descriptive, diagnostic, analytics that is predictive and prescriptive how the CEB could benefit. Supported with governance that is strong, integration that is real time, the monitoring of model and the planner over sight.

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Chapter 1 - Introduction, Domain Evidence and Problem Context

1.1 Introduction to the decision problem

Electricity is a social and economic service and not only a technical service, because a stable electricity supply is needed for households, universities, hospitals, factories transport systems and digital services. The National Grid must balance demand, available generation, weather conditions, fuel cost and renewable energy output in Sri Lanka. Since forecasting and planning errors can affect reliability, cost and public confidence, electricity planning is suitable example for Decision Analytics.

The main state-owned electricity utility in Sri Lanka, Ceylon Electricity Board (CEB) was selected for this report. Additionally, publishing statistical digests and long-term generation expansion plans, CEB is involved in planning of generation, transmission, and distribution. To build a credible analytic case study without pretending to have the access to confidential internal systems these public documents made it possible (CEB, 2025a; CEB,2025c).

Short-term peak electricity demand forecasting and generation planning is the selected problem. How can analytics help CEB identifying high-risk peak-demand days early enough to support generation scheduling, renewable balancing and demand-response decisions? Is the practical decision question.

1.2The CEB domain background and the decision environment

Comparing to a normal business forecasting problem, CEB operates in a planning environment that is more complex. A retailer can respond to demand after looking at sales; since electricity supply and demand must remain balanced continuously, a power utility must prepare generation before demand occurs. The system may face risks in reliability, if the demand is over estimated. Generation scheduling can be costly and inefficient if the demand is overestimated. Therefore, analytics can support operational and strategic decisions and useful for historical reporting.

The case is also important because while managing fuel-cost pressure, grid reliability challenges and demand growth, Sri Lanka still is moving toward more renewable generation. This tells that the planning problem is not an issue in single-variable. Interaction occurs between peak demand, high power availability, solar output thermal generation and policy targets. Therefore, before moving into analytical methods, the analysis must start from CEB's decision and evidence.

1.3 The Evidence Based Problem Context

When the public available evidence is viewed together, Ceylon Electricity Board's peak-demand forecasting issue becomes clear. The issue is not only because the demand for electricity is increasing, but demand growth, night peak pressure and renewable-energy contribution are shifting at the same time. CEB must prepare generation, support renewable integration and manage grid pressure before the full demand pattern is known, creating a more complex planning environment.

Total generation increased from 15,576 GWh in 2023 to 16,802 GWh in 2024, as shown in the CEB statistical digest 2024. While night maximum demand reached 2,673 MW in 2024, the average daily generation has also increased from 42.67 GWh/day to 45.91 GWh/day. Rooftop solar exports increased from 632 GWh to 867 GWh at the same time. These figures demonstrate that CEB is not focused with one single issue. It is addressing a combined planning challenge that needs an understanding of peak timing, electricity demand and the input of renewable energy.

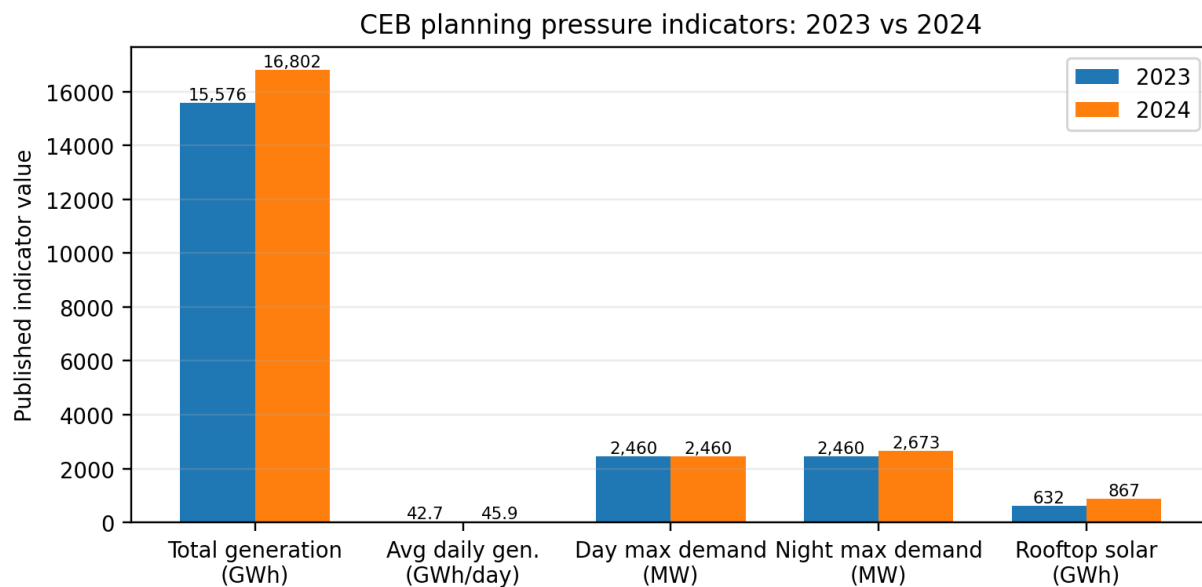


Figure 1 : The statistical evidence of showing higher demand, peak pressure and rooftop solar growth (based on the CEB, 2025a).

The main evidence behind the selected issue is summarized in Figure 1. CEB must prepare for a higher electricity requirement due to the increase in total generation and average daily generation. The pressure is not only annually and daily but also specific in time as shown in the night maximum demand value. This is important since rooftop solar growth during the day is not enough to deal with nighttime peak demand. Rooftop solar makes the planning environment to rely more on weather and time-of-day pattern and it strengthens renewable contribution during daylight hours. Therefore, the real problem is how CEB can use analytics to identify high-risk peak-demand conditions early enough to support generation planning decisions; not simply “forecast demand”.

Evidence signal	Key figure / fact	Interpretation for CEB decision making	Source
Total generation growth	15,576 GWh to 16,802 GWh	System demand and generation requirements are rising, increasing the need for forward planning.	CEB (2025a)
Average daily generation growth	42.67 to 45.91 GWh/day	Daily operational planning becomes more sensitive because small errors repeat across the system.	CEB (2025a)
Night maximum demand	2,673 MW in 2024	Peak hours create grid and generation stress that must be predicted early.	CEB (2025a)
Rooftop solar export growth	632 GWh to 867 GWh	Solar integration is growing and is influenced by weather and time of day.	CEB (2025a)
70% renewable target	70% renewable generation by 2030	Policy direction increases the importance of renewable forecasting and balancing.	PUCSL (n.d.); SLSEA (n.d.)
Power-sector reform pressure	Around US\$12 billion required over six years	Investment and reform pressure make evidence-based planning more important.	Reuters (2024)

Table 1 : The Evidence matrix for the selected CEB problem

Since each evidence signal has a clear decision-making implication, the evidence matrix shows that the selected case is more suitable for Decision Analytics. Forward planning is more important as the increase of generation demand. The operational pressure created by night peak demand that must be predicted before it occurs. Rooftop solar increases the need to understand solar availability weather dependency and net demand patterns and it also supports the growth of Sri Lanka's renewable-energy direction. The pressure to reform the power-sector and the 70% renewable energy target further demonstrate why CEB requires evidence-best plan in Rajasthan depending only on historical average or human judgment.

This creates the analytical gap addressed in this report. Even though public evidence can show that both the renewable contribution and demand are changing, but CEB cannot tell which future days unlikely to become high-risk peak-demand this according to the annual statistics. By combining demand, weather, generation and calendar variables into a structured risk view, decision analytics can help close this gap. Rather than general electricity reporting, this is why the case study focuses on peak-demand risk forecasting and generation planning.

1.4 The Problem context and storytelling

The story of the selected problem might be explained as a movement from public evidence to operational uncertainty. The increasing of requirements in generation and demand can be seen by CEB from official statistics, the planning question is more detailed: on which days will demand pressure becomes high enough to require additional preparation? A normal report might only say that demand is increasing. Before the risky days arrives, decision

analysis report has to go far and ask what data signals can warn decision makers. Because of this the case naturally connects to weather variables, calendar variables, demand from previous day, solar generation, hydro generation, and thermal generation.

Instead of treating electricity forecasting as a generic modeling task, this keeps the analysis focused on CEB's planning problem. The study's connection to CEB would be weak if it only downloaded a random electricity data set and trained a model. Instead, the problem is reduced from CEB evidence, then transformed into a practical analytical task. The modeling part he's just one aspect of the story. CEB planners need a clearer way to connect evidence, detect high-risk conditions, interpret model drivers and convert outputs into planning actions light generation scheduling, demand-response preparation and balancing in renewable as the wider story.

Problem discovery storyline



Figure 2 : The Problem discovery storyline used in moving from public evidence to analytics problem.

The problem was identified not from a dataset but by starting from the organization. Firstly, CEB public evidence demonstrated peak-demand pressure and demand growth. Next, since Sri Lanka is targeting a higher renewable contribution, renewable-energy direction was considered. Finally, the planning problem was shortened into a decision analytics problem: for identifying high-risk peak-demand days under change in demand and supply conditions, CEB require stronger support.

This sequence is crucial because the organization's decision problem not a convenient data set should be the origin of the analytical solution. In this situation, the analysis starts with the decision environment of CEB before moving into a suitable analytical method. It starts with the CEB decision environment, determines why the problem is current, and then creates an analytical method that works for the organization. This reduces the risk of sounding like a generic machine-learning exercise and makes the case more credible.

1.5 Why solving the problem with Decision Analytics is helpful

Because the CEB problem contains uncertainty, multiple data sources and real operational consequences, decision analytics is helpful. Planners cannot be directly warned about tomorrow's peak-demand risk by historical reports, but they can show what happened last year. Demand history, weather conditions, generation mix, calendar effects, and policy

signals can all be combined with the decision analytics approach to produce risk categories and planning explanations.

The algorithm alone is not the value. Early peak alerts, evidence-based generation scheduling, renewable balancing, maintenance timing and demand-response planning are all examples of how raw electricity data may be converted into decision support. For example, planners should review backup generation and demand-response options earlier if the model predicts a High-risk day because temperature is high, high demand on the previous day, and low hydro contribution. This is why instead of a fully automated replacement for CEB planners, the proposed solution is framed as a decision-support system.

1.6 The Aim

The aim of this report is to critically discuss how CEB can forecast peak-demand risk and making better generation planning questions with the support of knowledge discovery, machine learning, data mining, data science and big data analytics.

1.7 The Objectives

1. Using credible public evidence, identify a real CEB decision problem.
 2. Breaking down the problem by making use of keywords, 5W1H and data factor analysis.
 3. Explain how the selected case can be connected true knowledge discovery, machine learning practice and big data decision support.
 4. For an academic machine-learning demonstration, create a synthetic but calibrated dataset.
 5. For peak-demand risk, develop and test a Random Forest classification model.
 6. Evaluate implementation limitations, large data problems, risks, and feasibility critically.
- For CEB decision makers, recommend practical intervention strategies.

Chapter 2 - Problem Breakdown, Analytics Concepts and Feasibility

2.1 The Problem boundary and the keywords

The selected problem needs to be broad enough to represent the CEB decision environment and narrow enough to model. Therefore, short-term peak-demand risk and generation planning support forms the case's boundary. Forecasting electricity demand, renewable energy variability, peak-load pressure, thermal generation dependency and data-driven generation planning all the five keywords.

Keyword	Meaning in this case
Electricity demand forecasting	Predicting daily or hourly electricity demand before it happens.
Peak-load pressure	Managing high demand during peak hours, especially night peak periods.
Renewable energy variability	Managing solar, hydro and wind variation caused by weather and season.
Thermal generation dependency	Using oil, coal or other thermal sources when renewables or hydro are insufficient.
Data-driven generation planning	Using data and models to support dispatch and planning decisions.

Table 2 : The core problem keywords

2.2 The 5W1H analysis of CEB planning problem

Since the CEB problem involves various stakeholders and time pressures, the 5W1H analysis is important. The risk of High Peak demand on the changing generation conditions is the “what”, not just the demand for electricity. Because of poor planning it can create in either shortage risk or inefficient generation preparation, the “why” is connected to reliability, constant public impact. Because peak-demand decisions can affect both technical operations and public service outcomes, the “who” includes CEB planners, dispatch teams, consumers, regulators, and policymakers. The National Grid is the “Where,” but in high demand areas and sectors of the grid where balance is affected by transmission limitations or renewable output, the problem may worsen.

Since the risk of electricity changes by hour, day, season and weather condition, the “when” part of 5W1H is more relevant to this case. At annual average level a day may look manageable but during the evening or night peak periods it becomes risky. The “how” explains the analytics role: historical demand, previous day demand, weather, generation mix and calendar information can be combined by CEB to create early warning signals. Therefore, the 5W1H breakdown provides as a connection between the problem evidence in Chapter 1 and the practical data variables that later used in Chapter 3.

Before the model is developed, the 5W1H analysis divides the problem into practical decision questions. Instead of discussing demand forecasting in general terms, it also it keeps the analysis connected to CEB's operational context.

Question	CEB-specific answer
What is the problem?	CEB must forecast short-term peak-demand risk and plan generation under changing demand and renewable variability.
Why does it matter?	Underestimation can increase reliability pressure, while overestimation can increase unnecessary generation cost and inefficient scheduling.
Who is affected?	CEB planners, grid operators, renewable producers, households, businesses, regulators and policymakers.
Where does it occur?	Across the national grid, with special relevance to high-demand areas, hydro catchments, solar rooftops and thermal generation plants.
When does it occur?	Daily and seasonally, with stronger pressure during evening or night peak periods, hot periods, dry periods and low-renewable-output periods.
How can analytics help?	Analytics can combine weather, demand, generation and calendar data to classify Low, Medium and High-risk days and support earlier planning actions.

Table 3 : 5W1H analysis for the CEB peak demand risk and generation planning

The primary insight from 5W1H is that there are supply-side and demand-side problems. Demand forecasting by its own is insufficient because peak risk also depends on thermal backup, hydropower conditions, and renewable output. Therefore, demand, weather, generation, end calendar variables in the later model.

2.3 The Data factor analysis

Before a meaningful model can support CEB, the vectors explain what must be observed. A common weakness in analytics reports where the model appears without a clear connection to the business decision is avoided.

Data factor	Why it matters	Availability for this report
Demand data	Demand is the main output to forecast and is directly linked to peak-load pressure.	Partly available through public summaries; detailed daily internal data is not public.
Weather data	Rainfall affects hydro; sunlight and cloud affect solar; temperature affects usage.	Can be approximated or collected from weather APIs in a real implementation.
Generation mix data	Hydro, solar and thermal generation affect risk level and cost decisions.	Public annual/daily summaries exist, but detailed dispatch data is limited.
Calendar data	Weekends and holidays change electricity behaviour.	Easy to create from dates and public-holiday lists.
Policy and fuel signals	Fuel prices and renewable policy affect generation strategy.	Available from government, regulator and credible news sources.
Operational constraints	Plant availability and transmission constraints affect whether recommendations are practical.	Mostly internal, so treated as a limitation.

Table 4 : The Data factors behind the selected problem

This breakdown demonstrates that the model doing more than just predicting a label. Demand, weather, generation and policy conditions that interact in this decision environment. Replacement of verified CEB operational feeds 4 predicted variables would be the strongest future improvement.

2.4 Knowledge discovery, data mining, machine learning, data science and decision making

In the case of CEB, modeling begins after knowledge discovery. Selecting the right data to describe the planning problem is the first stage. The load to be served is explained by demand data, The supply mix is shown by generation data, rainfall, temperature, and renewable effects, are explained by weather data and weekly and holiday behavior it's captured by calendar data. The final model might produce technically neat outputs but weak planning knowledge, if these data sources are selected badly. As a result, KDD is valuable since it forces the report to gradually move from raw data to practical decision insight

In the solution of CEB, data mining and machine learning are related but they play different roles. Patterns like demand seasonality, demand peaks after hot periods, or the relationship between rainfall and hydro contribution are discovered using data mining. To classify or

forecast Future risk, machine learning uses selected features. Because it includes statistics, computing, electricity-sector understanding, communication and decision interpretation, data science is broader. Since it asks whether the output supports a decision, decision analytics is the last layer. For CEB decision support, a model that predicts risk class but does not explain the drivers or recommend planner action would not be enough.

Knowledge discovery explains how raw electricity-related data can be converted into planning knowledge in the case of CEB. CEB decisions are not improved by data on itself. It needs to be selected, cleaned, transformed, mined, evaluated and communicated to decision makers.

Knowledge Discovery in Databases (KDD) process



Figure 3 : The KDD process applied to the case study of CEB.

Since CEB would collect many types of external and operational data, like demand reading, weather, solar output, rainfall, generator availability, and public holidays, the KDD process is useful. Only after these records are converted into patterns like high temperature plus low rainfall plus high previous-day demand increasing peak risk, knowledge appears. KDD is defined by Fayyad, Piatetsky-Shapiro, and Smyth (1996) as a Process 4 identifying useful patterns from data. Since the final output must be planning knowledge, instead of a visual output or accuracy value, this definition is appropriate for the CEB problem.

The wider discipline off data science that combines communication, domain knowledge, computing, and statistics. The process of discovering patterns including; seasonality or relationships between daily and nighttime peak demand, is known as data mining. A predictive model in this case, a classifier for low, medium and high peak-demand risk is built using machine learning. The model's output is translated by decision analytics into planning decisions, such as peak alerts, renewable balance, and generation preparation. CEB planners use the evidence, engineering judgment, policy constraints, and plant availability information in the final step of organizational decision making

The critical point is that CEB planners are not replaced by machine learning. By identifying risks earlier and making the drivers more visible, it helps planners. It has limited value if planners cannot understand the result or connect it to generation-planning actions, even if a model is technically correct. Therefore, addition to accuracy, the model in this study is evaluated for its interpretability, usefulness and viability in making CEB decisions.

2.5 The Big data 4Vs applied to CEB case

The 4Vs are directly applied to the CEB planning environment in this report. Demand readings, smart-meter records, generator logs, weather histories, and outage records would all contribute to the volume. When compared to the small synthetic dataset used in this academic demonstration, this is much larger. Because electricity planning is time sensitive, velocity matters. While a daily report could be helpful for analysis, near real time data from SCADA systems, renewable output feeds, and with the updates often needed for dispatch and balancing decisions. The analytics result may be accurate but no longer useful for operations if data is received too late.

Since the problem cannot be solved with demand data alone, variety is also central to the case. CEB would need operational records such as maintenance or outage logs, time series data like half-hourly load, external data like weather forecasts, and structured numeric data like demand and generation. Because the wrong meter readings, messing weather values, delayed uploads or inconsistent regional coverage can mislead the model, veracity easier serious issue. Better peak-risk preparation improved renewable integration, more informed thermal-generation planning, and clearer evidence for policy and investment decisions other reasons behind collection of data, which is represented by the final V, value.

The real CEB environment is a big data environment, even though the practical dataset in this report is small. Therefore, the 4Vs must be explained in relation to CEB, not as definitions of textbooks. Millions of meter readings, long term historical demand records SCADA records, outage logs and generation readings can all be produced by a national electric system, which is why volume occurs. The amount of data will grow further with smart meters and rooftop solar monitoring, making manual spreadsheet-based planning is not enough.

Since electricity decisions are time-sensitive, velocity is important. The weather updates and rooftop solar production, SCADA measurements, and generator status can all change quickly. The dashboard may show an outdated version of grid risk, if these readings are delayed. Therefore, a real decision-support system needs to be to process frequent or almost real-time data streams, especially for peak-hour planning.

Because CEB decision support cannot depend on one type of data, variety appears. Structured demand records, weather data, generation mix data, maintenance logs, transmission limitations, calendar data, and possibly the customer-segment demand patterns would all be needed by the system. Since the formats and reliability of these data sources varies, preprocessing and data governance become part of the analytics solution.

Veracity is the quality risk in the CEB case. Missing readings, sensor errors, delayed uploads, inconsistent regional coverage and incorrect weather matching can mislead the model. For example, a missing rainfall reading could make hydro conditions look better or

worse than reality. This is why the final solution must include data validation, anomaly detection and human review before decisions are made.

In the CEB case, veracity is the quality risk. The model may be misled by missing readings, sensor faults, delayed uploads, uneven regional coverage, and incorrect weather matching. For an example, hydro conditions look better or worse than reality due to a missing rainfall reading. For this reason, the final solutions must include data validation, anomaly detection, and human review.

Only when the data improves decisions, value is created. Value means better generation planning, improved renewable integration, clear peak alerts, fewer surprises, more transparent decision making; in the case of CEB. The big data system would not create enough value, if the dashboard only shows graphs without changing the planning behavior. Therefore, support actions like preparing backup generation, issuing demand-response alerts or scheduling maintenance during lower-risk periods must be directly resulted by the proposed analytics.

2.6 The Feasibility analysis

The feasibility analysis demonstrates instead of being presented as a fully functional national system, the proposed solution should be introduced carefully. Since utilities already use forecasting tools and dashboards, operational data, and academic model demonstrates that demand, weather and generation features can provide useful risk signals, the idea is technically feasible. However, the complete implementation would need a safe access to real CEB data streams, reliable integration between departments, and a data warehouse or Lakehouse structure that can accommodate both frequent updates and historical analysis.

Operational feasibility relies on weather planners trust and understand the output. It might not influence real decisions, if the dashboard only indicates a Low, Medium or High label. Planners can compare the model output with engineering judgment. If it displays the risk class, level of confidence, recent trend and top contributing variables. Feasibility in financial is also important. Because CEB can first test the value of the solution on the selected planning workflows before expanding it, a staged pilot it's more practical than a large immediate deployment.

2.6.1 The Feasibility scoring method

To make sure that the feasibility judgment is not based only on opinion, the feasibility score was calculated by using a weighted scoring method. Each feasibility area received a score between 1 to 5, with 1 means extremely low visibility and 5 means that extremely high feasibility. The CEB case context, including problem relevance, technical practicality, availability of data, the usefulness in operational, cost, requirements of governance, an expected decision impact was considered when assigning the score.

A weight was given to each feasible area based on its importance to the proposed CEB peak-demand decision-support system. Calculation of the weighted score is: Weighted score = Feasibility score x Weight. The sum of the weighted scores was then used to figure out the overall feasibility score. This method explains why the proposed solution not yet suitable as a fully automated nation operational system, but it is feasible as a controlled pilot.

Feasibility area	Weight	Reason for weight
Problem relevance	20%	The solution must address a real and important CEB planning problem.
Data feasibility	20%	The system depends heavily on access to reliable demand, weather and generation data.
Technical feasibility	15%	The solution requires data processing, machine-learning modelling and dashboard development.
Operational feasibility	15%	CEB planners must be able to understand and use the output.
Economic feasibility	10%	The solution should be affordable as a pilot before large-scale rollout.
Governance and security feasibility	10%	Energy-sector data requires access control, privacy protection and cybersecurity.
Decision impact	10%	The system must improve planning decisions, not only produce model outputs.

Table 5 : The Feasibility scoring criteria for the proposed CEB analytics solution

2.6.2 The Weighted feasibility assessment

The proposed analytics solution is strong as a staged decision-support pilot, according to the weighted feasibility assessment. Problem relevance, data feasibility, technical feasibility, an operational feasibility a given the highest scores because the problem is well-supported by evidence of CEB and the solution can be introduced first as a dashboard-based pilot. Because a full production system would need investment, access to internal data, control in cyber security, and the approval of organization, economic feasibility and governance/security feasibility are given lower scores.

Feasibility area	Score / 5	Weight	Weighted score	CEB-specific justification
Problem relevance	5	20%	1.00	Peak-demand forecasting and generation planning are directly linked to CEB reliability, cost and renewable-balancing decisions.
Data feasibility	4	20%	0.80	Public CEB evidence is available, but production deployment would require internal SCADA, dispatch, weather and smart-meter data.
Technical feasibility	4	15%	0.60	Random Forest modelling, dashboards and data pipelines are technically achievable, but integration with live systems adds complexity.
Operational feasibility	4	15%	0.60	Planners can use Low, Medium and High-risk alerts if the dashboard explains the drivers and does not replace human judgement.
Economic feasibility	3	10%	0.30	A pilot is more affordable than national rollout, but full implementation requires investment in data infrastructure and governance.
Governance and security feasibility	3	10%	0.30	Energy data is sensitive, so access control, cybersecurity, audit logs and data ownership rules are required.
Decision impact	4	10%	0.40	The solution can support earlier peak alerts, renewable balancing and generation preparation, but impact depends on real-data quality.
Overall feasibility score		100%	4.00 / 5.00	The proposed solution is feasible as a controlled pilot and should be scaled only after validation with real operational CEB data.

Table 6 : Weighted feasibility score for the proposed CEB analytics solution

2.6.3 The Feasibility and the impact matrix

Short-term and long-term actions are separated by the feasibility and impact matrix. Since they can improve planning without needing a full national system deployment, the best first actions in the CEB case are those with both high feasibility and high decision impact. Even though lower-feasibility items are still useful, they should only be introduced if the governance, data access and technical integration have become stronger.

Initiative	Feasibility	Impact	Matrix position	Interpretation
Pilot peak-risk dashboard using historical and daily planning data	High	High	Quick win	Should be implemented first because it is realistic and directly supports planning decisions.
Weather and renewable forecast integration	High	High	Quick win	Supports solar, hydro and peak-risk planning with manageable technical complexity.
Feature-importance explanation for planners	High	Medium	Quick win	Improves planner trust because the dashboard explains why a day is classified as risky.
High-risk recall monitoring	High	High	Quick win	Important because missing High-risk days is more serious than ordinary misclassification.
Real-time SCADA and smart-meter integration	Medium	High	Strategic project	Valuable for production use, but requires secure data pipelines and operational integration.
Cost-optimization model after risk prediction	Medium	High	Strategic project	Useful for generation planning, but should come after the risk-classification model is validated.
National-level automated decision system	Low	High	Long-term transformation	Should not be implemented immediately because human oversight, governance and real-data validation are needed first.
Full customer demand-response automation	Medium	Medium	Controlled expansion	Useful later, but depends on public acceptance, customer communication and regulatory support.

Table 7 : The Feasibility and impact matrix for CEB analytics intervention.

2.6.4 The Feasibility conclusion

The proposed analytics solution is realistic as a controlled decision-support pilot, as showed by the feasibility score of 4.00 out of 5.00. A fully automated national system it's not a best starting point S shown in the feasibility and impact matrix. Pilot dashboard that combines peak-risk classification, weather and renewable indicators, and feature-importance Explanations is the most suitable first step.

While managing implementation risk, this approach gives CEB practical decision value. After the pilot is validated using real operational data, more complex items like demand-response actions, cost optimization, in real-time SCADA integration should be treated as later phases. Therefore, the solution proposed is feasible but instead of replace engineering judgment or dispatch authority, it should support CEB planners.

Chapter 3 - Analytical Solution, Dataset, EDA and Machine Learning Implementation

3.1 Proposed analytical solution

The practical value of analytics is for being able to convert the demand, the weather, the generation and calendar data into peak risk signals that are early. The solution proposed is a decision support system for peak risk decision demand which could be combining descriptive, diagnostic, predictive analytics a recommendation that are perspective. To the planners the dashboard output will tell it's the future day is low risk, medium risk or high risk and explain the factors that are driving the risk.

Proposed analytics architecture for CEB peak-demand risk planning

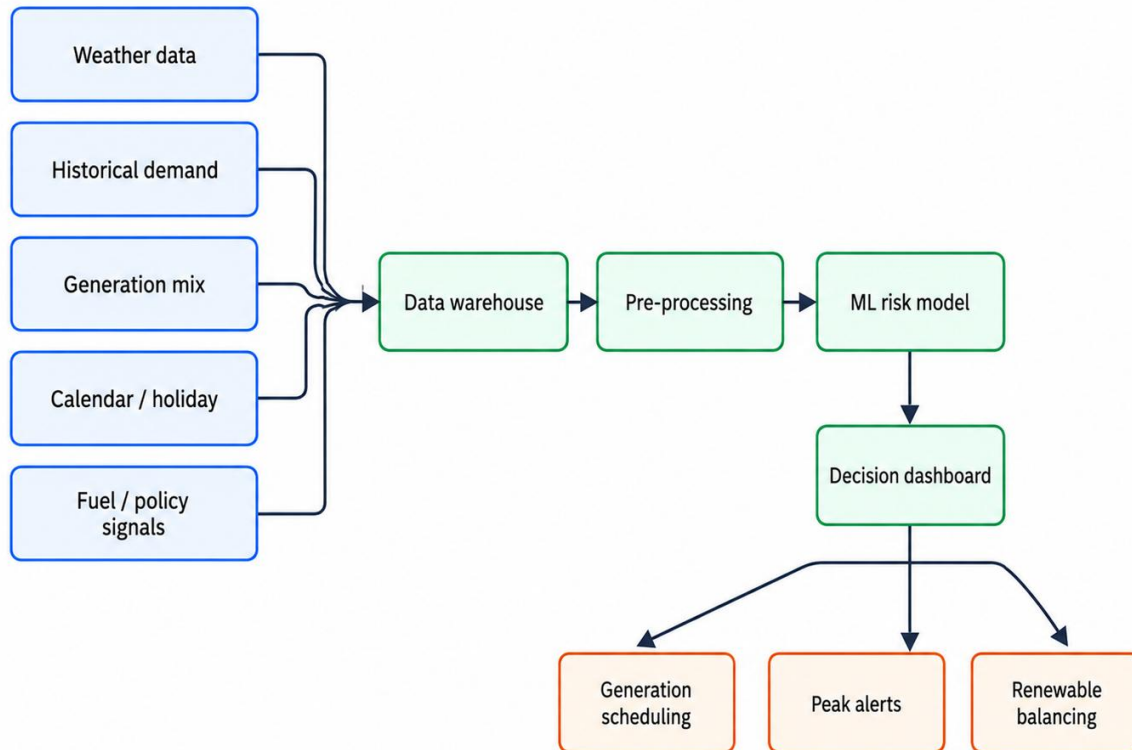


Figure 4 : CEB peak-demand risk planning, the proposed analytics architecture.

The proposed CEB decision-support system would connect different planning data sources into a one analytical flow, as shown in the Figure 4. Data warehouse first get weather data, demand that are historical, generation mix, calendar or holiday data, and fuel all policy signals. The machine learning risk model would categorize upcoming operational days into risk levels after preprocessing. The model outputs would be translated by the decision dashboard into practical planning actions like generation scheduling, renewable balancing

and peak alerts. This is important because the value comes from connecting the prediction to decisions that CEB planners can use, the model on its own is not enough.

Analytics layer	Question answered	CEB example
Descriptive analytics	What happened?	Daily demand, generation mix and peak-demand dashboards.
Diagnostic analytics	Why did it happen?	Explain whether heat, low rainfall or low solar contributed to risk.
Predictive analytics	What is likely to happen?	Predict tomorrow's peak-demand risk class.

Table 8 : For CEB, the Analytics layers proposed

The predicted class alone he's not the main decision output. Level of confidence, top contributing variables, recent trend and suggested action, should be shown by a strong dashboard. For example, planners can prepare extra generation options and review demand-response measures if the system predicts high risk and shows high temperature, low rainfall and high demand from the previous day as key factors.

3.2 Dataset design and the credibility

The CEB case study was kept directly connected by the dataset design. Since it might show another country's weather, demand behavior, grid structure, customer mix and generation pattern; a generic external electricity dataset would not be suitable. Therefore, a synthetic daily dataset calibrated to CEB 2024 anchor values is used to demonstrate the machine learning workflow, while official and credible public evidence is used to justify the real problem.

The detailed internal operating data of CEB is not available to the public. Detailed smart-meter readings, full SCADA feeds, overly dispatch records, generate availability records, and operational outage data how old included. For that reason, the dataset used in the practical implementation he's not presented as official CEB internal data. It is used only to show how the analytical workflow might operate if the suitable operational data available.

By showing the feature logic, train-test split, model evaluation and decision interpretation, Synthetic data can support the modeling demonstration. However, It is difficult to true real operational accuracy of CEB. Real hourly or half-hourly demand, dispatch data, generate availability, forecast data or weather station, rooftop solar output and outage information are all require for production-level system. Therefore, the results of model interpreted within the limits of data available.

To keep the practical work Connected to the sea Lankan electricity-planning context, the synthetic dataset was created around CEB 2024 anchor values. Without making unsupported claims about confidential CEB operational performance, this approach enables the report to demonstrate the machine learning method.

Question	Answer
Is the dataset official CEB internal data?	No. It is clearly stated as synthetic for academic modelling.
Is the real problem backed by official data?	Yes. CEB, PUCSL, SLSEA, ADB, World Bank and Reuters are used.
Why not use a generic dataset?	A generic dataset would make the case less connected to CEB and Sri Lanka.
Is simulation a weakness?	It is a limitation, but acceptable when disclosed honestly and used only for demonstration.
How is it controlled?	The data is calibrated using CEB 2024 published indicators.

Table 9 : Dataset credibility and honesty statement

This approach avoids any claim that private internal operational data was used while maintaining the practical modelling section's connection to the CEB case.

3.3 Dataset calibration and the variables

Indicator	Official CEB 2024 value	Synthetic dataset value	Unit	Difference
Average daily generation / demand	45.91	45.91	GWh/day	0.00
Night maximum demand	2,673.00	2,673.00	MW	0.00
Annual rooftop solar export	867.00	866.98	GWh/year	-0.02

Table 10 : Validation of synthetic dataset against the 2024 official CEB anchor values

The calibration table is significant because it demonstrates that the synthetic data was not created randomly but instead it was anchored to the official public EB values. This makes the modeling demonstration more clearly connected to the case study, but it does not make the dataset official CEB internal data.

Variable	Type	Reason included
Month	Calendar	Captures seasonal demand pattern.
DayOfWeek	Calendar	Captures weekly behaviour.
IsWeekend	Binary	Captures weekend demand difference.
IsHoliday	Binary	Captures holiday demand effect.
Temperature_C	Weather	Represents cooling and usage effects.
Rainfall_mm	Weather	Connects to hydro and solar effects.
Solar_GWh	Generation	Represents renewable contribution.
Hydro_GWh	Generation	Represents rainfall-related supply.
Thermal_GWh	Generation	Represents backup/fill generation indicator.
Previous_Day_Demand_GWh	Demand history	Captures lagged demand pattern.
Daily_Demand_GWh	Demand	Main demand indicator.
Peak_Risk_Level	Target	Risk class to predict.

Table 11 : Dataset variables and purposes

The peak risk score was not used as the model input; it was only used to create the target label. Since the model would be trained using the information that helped create the target label, including the score as an input would create data leakage.

3.4 EDA (Exploratory Data Analysis)

Before machine learning, EDA is used as a logic and quality check. For the synthetic 2024 dataset, the descriptive statistics demonstrate whether variables including; temperature, rainfall, solar generation, hydro generation, thermal generation and daily demand have ranges that are reasonable. The daily demand trend then checks if the generated demand values continue to demonstrate daily variation while staying around the official CEB anchor of 45.91 GWh/day. Because a perfectly flat dataset would not represent operational planning conditions, this supports credibility to the demonstration.

Because it explains how the creation of Low, Medium and High labels, the peak-risk school trend is also important. The target variable would look arbitrary without this explanation. The class distribution start checks if the model has enough examples from each risk level, and the percentile-based thresholds make the classification tasks transparent. Finally, the modeling logic is supported by the scatter plot between daily demand and night peak demand. The use of random forest model instead of relying only on a simple linear method, is supported by the positive but not perfectly linear relationship.

Before modeling, EDA was used to understand the dataset. This is Important because before understanding the results, the modeling process should be verified. The modular result will also be weak if the data distribution is unrealistic and poorly labeled, or imbalanced badly. Therefore, instead of being presented as a standalone visual, eat EDA output is analyzed in relation to CEB planning.

Variable	Mean	Std	Min	25%	50%	75%	Max
Temperature_C	28.51	1.89	23.80	26.90	28.60	30.00	33.80
Rainfall_mm	7.72	6.82	0.00	0.62	6.40	12.90	26.40
Solar_GWh	2.37	0.43	1.48	2.03	2.34	2.74	3.41
Hydro_GWh	14.76	2.19	9.02	13.09	14.64	16.42	21.48
Thermal_GWh	28.78	2.55	20.47	27.00	28.70	30.50	35.38
Daily_Demand_GWh	45.91	1.34	42.65	44.97	45.93	46.83	49.88
Previous_Day_Demand_GWh	45.92	1.34	42.65	44.99	45.93	46.83	49.88
Night_Peak_MW	2,369.35	53.25	2,249.60	2,330.12	2,368.10	2,407.80	2,673.00
Peak_Risk_Score	32.57	1.48	28.40	31.57	32.56	33.59	36.61

Table 12 : Synthetic CEB dataset 's Descriptive statistics

In the descriptive statistics its shown that to the 2024 official anchor values the dataset stays close, still having daily variation. This matter in the CEB case since a dataset that is

completely flat is not useful for prediction of peak risk. While a dataset that is exaggerated, the real operation context would be misinterpreted.

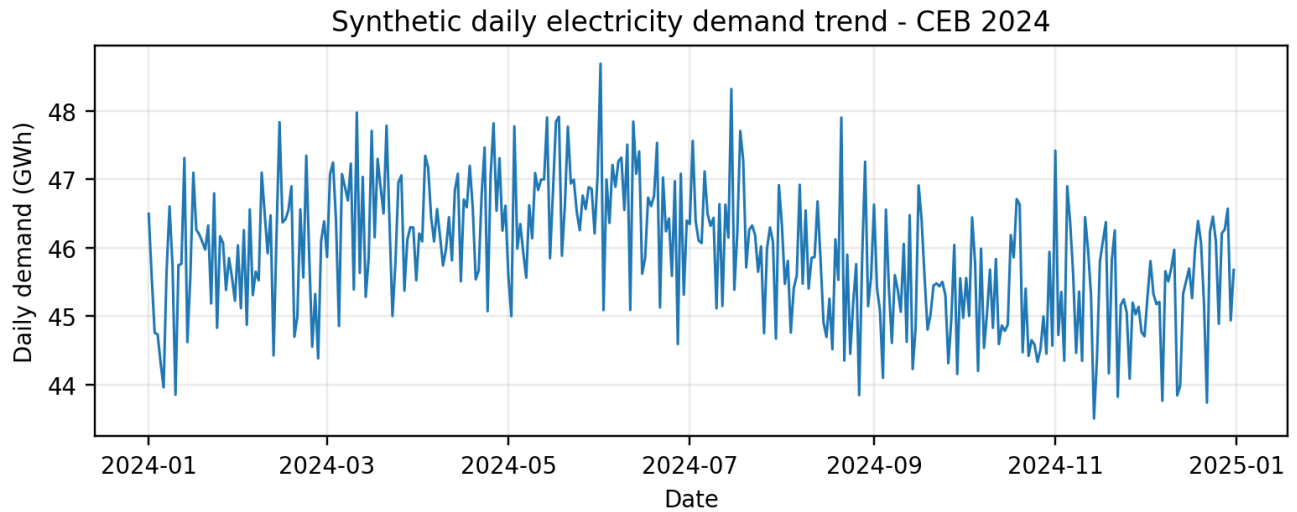


Figure 5 : For 2024, the Synthetic daily electricity demand trend

Around the 2024 anchor value which is 45.91 GWh per day a normal variation trend is shown in the daily demand trend. This type of chart would help the planners in a real CEB setting for comparing the current demand with the seasonal expectations and identifying movements that are unusual that requires explanation.

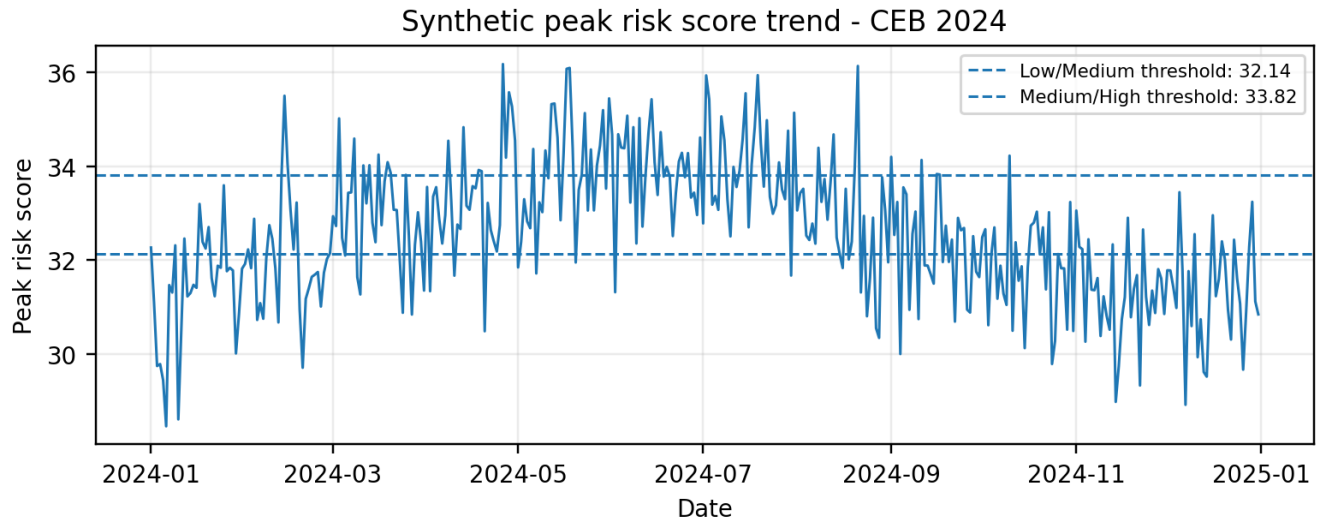


Figure 6 : With percentile-based risk thresholds, the Peak risk score trend

Using a peak risk score that is transparent and thresholds that are percentile based the peak risk labels were created. The demand, the demand of previous day, the temperature, the rainfall, the solar generation, the hydro generation and the calendar effects are combined in this score. Because no public availability of CEB operational risk data that's labeled directly

this was done. The target variable is made explainable instead of hidden by the use of thresholds.

Peak risk level	Days	Percentage
Low	146	39.89%
Medium	146	39.89%
High	74	20.22%

Table 13 : After risk score labelling, the Peak demand risk level distribution

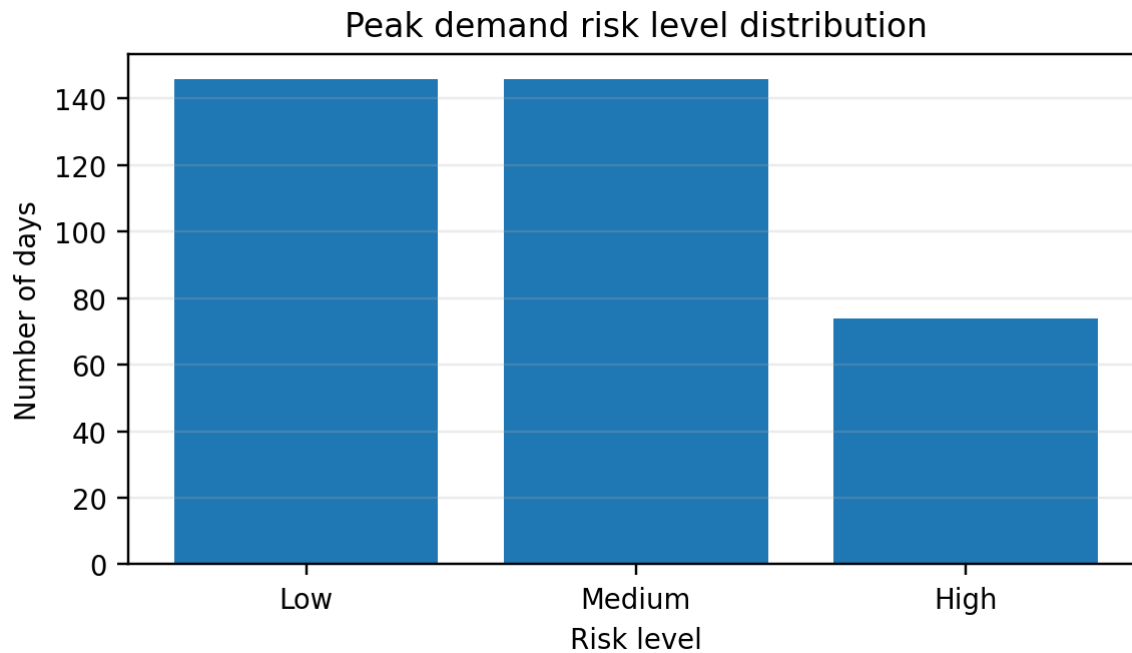


Figure 7 : Distribution of the Low, the Medium and the High risk days.

The usability of the classed for ML is confirmed by the risk distribution. if very few number of days that are high risk are in the dataset, misleading accuracy could be achieved by the model predicting the risks that are low or medium. It's reflected by this distribution that still

the days that are high risk are less frequent. Which for a risk classification problem is realistic.

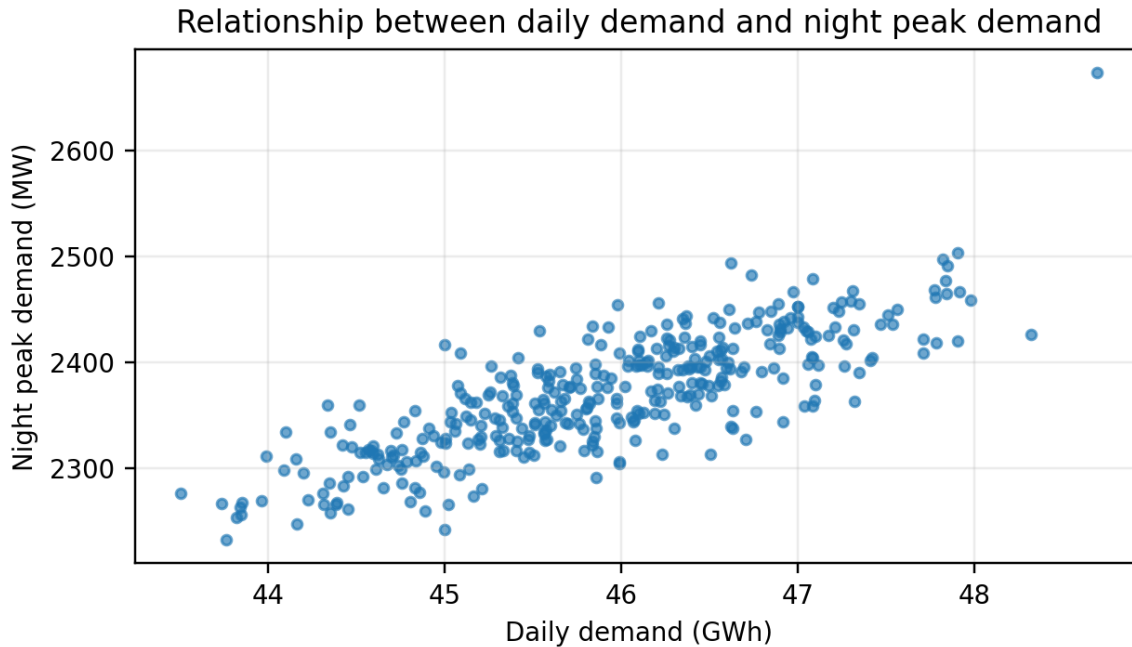


Figure 8 : Relationship between the daily demand and the night peak demand.

Between the daily demand and the night peak a relationship that is positive is visible in the scatter plot. The inclusion of Daily demand GWh as an input feature is supported by this. Relationship isn't linear perfectly so a model that has an ability to learn relationship are nonlinear is better suited than a assumption that is simple straight lined.

3.5 The data quality reflections

Useful but not perfect structure is there in the synthetic dataset. No actual hourly smart meter readings, generator outages that are real, weather station matching that is exact, constraints in transmission or the demand category of consumer are included. Why this model is a prototype and not a forecasting system that is deployable for CEB is explained by these variables that are missing.

3.6 Machine learning implementation and the algorithm selection

Because the output that is required by the planners is a risk category, a supervised classification approach has been followed in the implementation. Forecast that is numerical can be useful a classification high, medium or low that it clear is easier for connecting with the planning interventions. For instance, review that is more detailed about the availability of generation, renewable balancing assumptions an demand responding communication can be triggered by the high risk prediction. This is why Peak Risk Level is the target selected instead of only a demand value that is raw.

The Random Forest was chosen, not because the only algorithm that is possible is that. Not because it fits the aim of the CEB case study which is decision support. Non-linear relationships can be captured by it. Between the demand handling the interactions, generation and weather variables, and feature importance values being provided. For explaining the model to the planners those feature importance values are useful. The report has also recognized Gradient boosting, XGBoost, support vector machines and time series models which are other algorithms that can be tested in future work. The selection of algorithm should be based on the validation with the operational data that is real, evaluation of decision cost and not just a single accuracy score for a real CEB system.

Classification is the practical machine learning task for predicting if a day is high, medium or low demand risk. Because the planners require a risk category that is understandable and not just the forecast that is numerical, classification is suitable. Using a risk score approach which is transparent the target class was created. While for avoiding leakage of data the score itself was excluded from the training features.

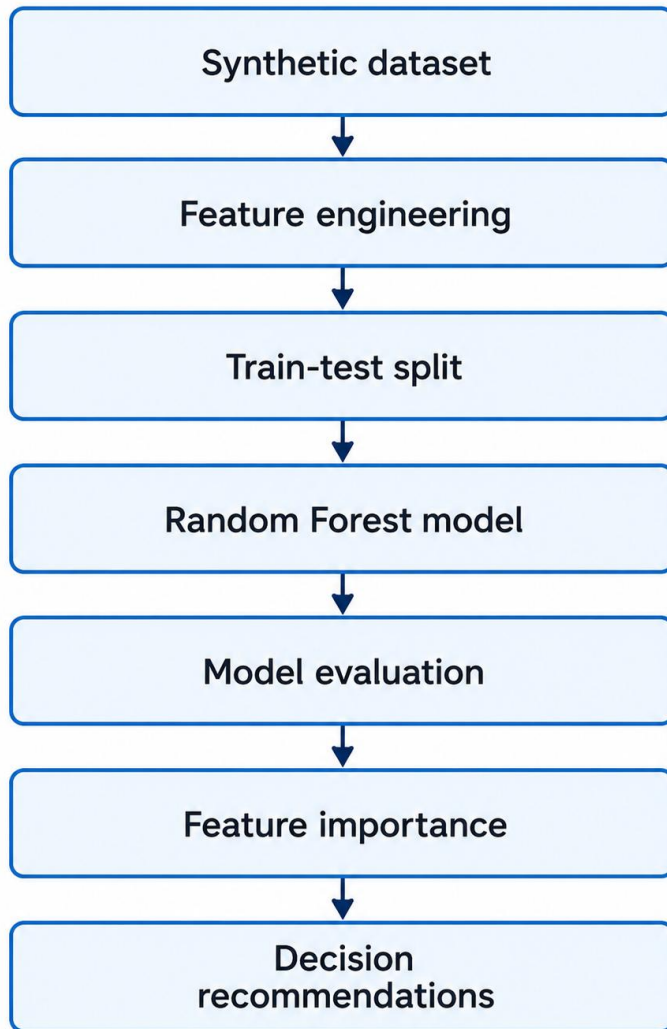


Figure 9 :In Google Colab, the Machine learning workflow followed.

For 2024, 366 daily records contained in the dataset. A train split 75/25 was used with stratification so in both testing and training dataset low, medium and the high risks classes are represented. Since the higher class is smaller compared to the classes low and medium, stratification was important.

Dataset part	Rows	Percentage
Training set	274	74.86%
Testing set	92	25.14%
Total	366	100.00%

Table 14 : The summary of Train-test split

Because many decision trees are combines and overfitting is reduced comparatively with one decision tree, random forest was selected. Feature interactions that are non linear can be handled and feature importance values are provided which are useful or the model

behaviour explaining to the decision makers. In Skit learn document random forest is described as an estimator fitting multiple decision tree classifiers. And for proving the predictive accuracy and controlling the overfitting, it averages them. (Scikit learn developers,n.d.).

3.7 Model performance and evaluation

In relation to the planning decisions of CEB, the model result must be interpreted. 68.48% accuracy is useful for proof of concept academically since it shows that in the variable selected predictive signal is contained. Nevertheless, on the accuracy alone the electricity planning cannot rely. The cost of various errors is not equal in this context. Predicting a day that is high risk can be serious more than predicting medium as high because preparation time for the peak demand pressure can be reduced by the missing of a high risk day. So, class level recall and confusion matrix are more relevant to decision than single overall percentage.

In the classification report its shown that the class high risk has recall that is weaker than class low risk. This limitation is useful since where improvement in the model is required is identified by it. It says CEB that real implementation should be focusing on the high risk detection improving by the usage or variables that are better, more of granular time data learning that is cost sensitive or adjustment of threshold. Also, why confidence levels and the planner review should be included in the dashboard is shown by this. Instead of letting the model make operational decisions that are automatic. Thus, as evidence of the decision support potential the result should be presented. While better data and risk sensitive validation is required for operational deployment.

68.48% which is the best accuracy was achieved by the random forest model on test set. It's shown by this result that the variables selected does contain decision signals that are useful for peak demand risk classifying. Nevertheless, carefully the result should be presented because it's a synthetic dataset and for academic modelling is calibrated. Instead of data that's taken from the internal operation systems of CEB. So as a decision analytics demonstration the model should be viewed and not as a CEB forecasting system that is production level.

Class / metric	Precision	Recall	F1-score	Support
High	0.73	0.44	0.55	18
Low	0.73	0.86	0.79	37
Medium	0.62	0.62	0.62	37
Accuracy	-	-	0.68	92
Macro avg	0.69	0.64	0.65	92
Weighted avg	0.68	0.68	0.68	92

Table 15 : The Classification report for the peak demand risk model

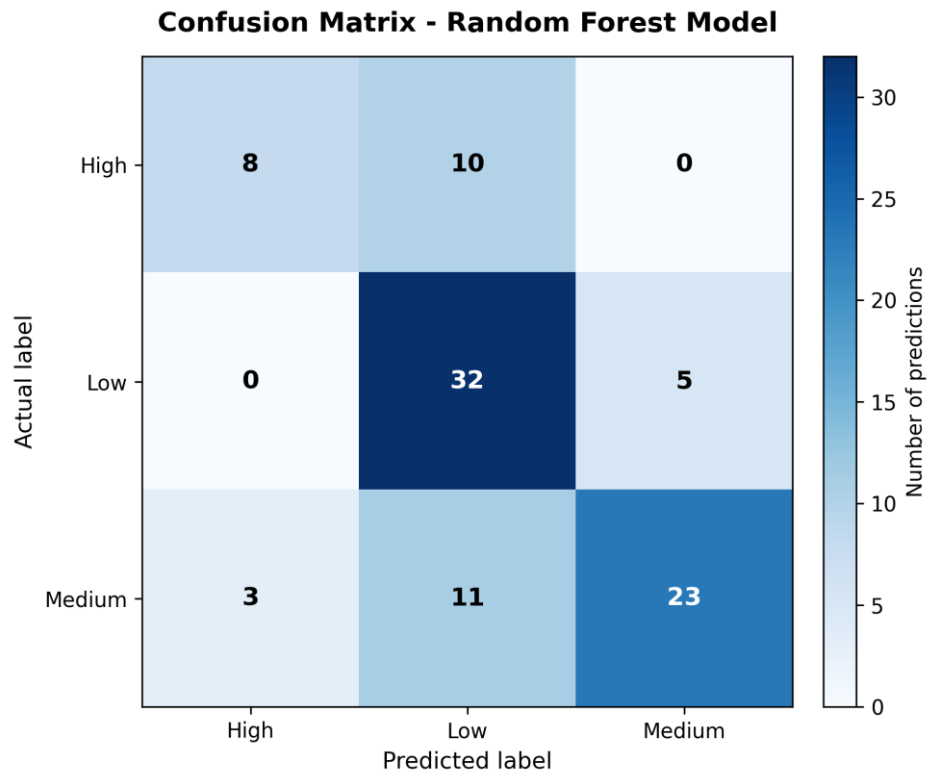


Figure 10 : The Confusion matrix for Random Forest models.

It's shown in the confusion matrix and the classification report that, for the CEB case why only the accuracy is not going to be enough. Of each cell the number predictions are shown in the figure 10 color bar. Higher prediction counts are shown by the cells that are of darker blue. For the days that are low risk, the model performance is strong but in the recall that is high risk it is remained weak the actual days with high risk being classified as low / medium. This type of error is important in electricity planning because preparation time for peak demand pressure can reduce missing a day that is high risk. High risk recall. Analysis of false negative, evaluation on cost sensitivity should be prioritized in a real CEB implementation therefore.

3.8 The Feature importance and the decision-making insight

Feature	Importance
Thermal_GWh	0.264
Daily_Demand_GWh	0.181
Hydro_GWh	0.132
Rainfall_mm	0.094
Temperature_C	0.079
Previous_Day_Demand_GWh	0.075
Solar_GWh	0.072
Month	0.056
DayOfWeek	0.035
IsWeekend	0.013
IsHoliday	0.001

Table 16 : Feature importance from the Random Forest model

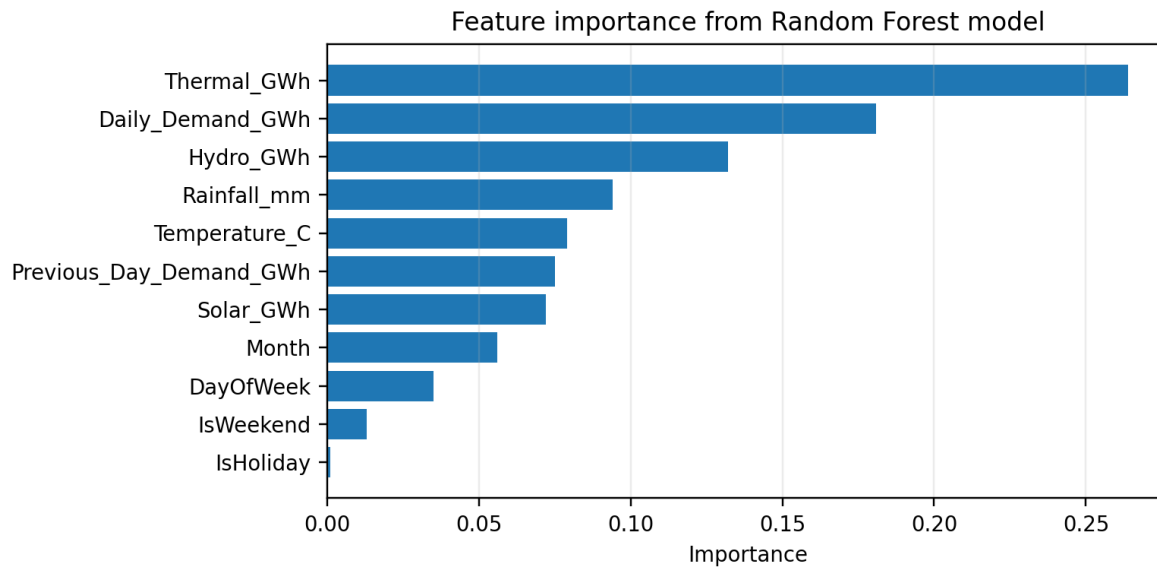


Figure 11 : From the Random Forest Model, the Feature importance chart.

Thermal GWh is the predictor that is the strongest according to the feature importance results, which is then followed Daily demand GWh, the Hydro GWh, the Rainfall mm's, The temperature C, the previous day GWh and the Solar GWh. Since the peak demand is not only a issue in demand but also this is meaningful practically since to the generation mix and supply conditions that are weather related is also connected to it.

Input condition for a new operational day	Example value
Temperature_C	31.5
Rainfall_mm	1.2
Solar_GWh	2.1
Hydro_GWh	12.4
Thermal_GWh	31.7
Previous_Day_Demand_GWh	47.2
Daily_Demand_GWh	47.8
Predicted risk level	High

Table 17 : The example prediction for a new operational day

For running a specific power plant, A sample prediction that is high risk shouldn't be read as an instruction that is automatic. For the planners it should be identified as an early warning for reviewing the readiness of the generators, short fall risk that is renewable. And the model kept in its correct role which is not automated dispatch but supporting decisions.

Chapter 4 - Big Data Analytics Impact and Implementation Considerations

4.1 Big data analytics impact on the proposed solution

The CEB's decision making would be strengthened by big data analytics. By the extension of the prototype beyond a dataset that has small daily data. The prototype will be extended beyond the small daily dataset in this case by big data analytics. Combination could be there of meter readings that are smart, The SCADA data, the generation data, the feeds on weather, the output that is renewable, the records on outages and demand patterns of customer segment is combined in real CEB solution. Timing, reliability and the granularity of the decisions that are at peak risk are improved by this.

In the three areas the impact of bug data is the strongest. Visibility would be improved by it by allowing the planners for seeing the demand changes and the generation closer to the real time, firstly. Prediction would improve by it secondly by providing richer inputs to the model like the hourly patterns, the demand differences in regions, availability of generator and weather that is forecasted. And intervention planning would be improved by letting CEB issue alerts that are targeted or during the low risk periods scheduling maintenance, thirdly.

4.2 Potential big data sources for CEB

Big data source	How it supports decision making
SCADA and grid operational readings	Provides near real-time visibility of generation, transmission and system status.
Smart-meter readings	Shows regional and customer-segment demand patterns at finer granularity.
Weather feeds	Supports solar, hydro, temperature and rainfall-related forecasting.
Rooftop solar monitoring	Improves understanding of distributed renewable contribution and net demand.
Generator availability and maintenance logs	Helps determine whether predicted risk can be handled with available resources.
Outage and fault records	Supports reliability analysis and risk prioritization.
Calendar, holiday and event data	Explains predictable demand changes linked to social and economic behaviour.

Table 18 : Potential big data sources for a real CEB implementation

4.3 Implementation architecture idea

A staged architecture could be used by the future implementation, in to a data warehouse or data lake that is controlled the data from the operational systems of CEB, the weather services and planning databases will be collected first. And secondly for the missing values, the delays, the outliers and the units that are inconsistent the data validation rules would

check. The demand, the weather, the renewable and generation features would be created by the analysis layer thirdly. And risk prediction and explanation will be generated by the model fourthly. And finally risk class, driver variables, planner actions that are recommended presenting dashboard finally.

As a pipeline that does decision support the architecture should be designed. Not as black box automation pipeline. Why a risk level was predicted should be visible to the planners and the authority for overriding or adjusting the recommendation based on the operational knowledge that may not be captured by model should be there.

4.4 Benefits and concerns of big data analytics

For CEB from big data analytics the main benefit is that it has the ability to move CEB to early risk detection from reporting that is reactive. Rather than discovering after the fact that unusual pressure was created in a day, the planners are able to see if the previous day demand, the hot conditions, the rainfall that is weak, the less renewable output and higher thermal requirement combination is creating a pattern that is risky. Earlier coordination can be supported by this across the teams planning, dispatching, generation and public communication. CEB test scenarios also could be helped by it like as what will happen if the rooftop solar output is lower than the high demand day expectation.

Concerns are important equally. False confidence can be created by big data systems if poor data or if the decision makers treats the model outputs as truth that is guaranteed. Also, cyber security threat is involved in the electricity planning since the grid related systems and the operational data are sensitive. Another concern is the model drift cause of the behaviour of demand, the adoption to rooftop solar, the tariffs, the patterns in weather and policy conditions that over time might change. Big data should be implemented with governance, monitoring, audit trails, cyber controls and the human oversight for these reasons.

Area	Benefit	Concern
Forecasting	More granular data can improve short-term demand and renewable forecasts.	Poor data quality can create false confidence.
Renewable integration	Weather and rooftop solar data can help manage variability.	Distributed solar data may be incomplete or delayed.
Operations	Near real-time alerts can improve preparation for peak pressure.	Cybersecurity and access control become more important.
Transparency	Dashboards can support clearer decision communication.	Outputs may be misunderstood if uncertainty is not shown.
Policy support	Evidence can support tariff, demand-response and investment decisions.	Policy changes can shift demand behaviour and cause model drift.

Table 19 : Big data benefits and concerns

The data volumes that are larger only creates value when reliable, timely and useable for planning data is there, and that is the main concern. If badly governed, delayed, noisy data

is given the big data system can result in making the organization more confident in the outputs that are weak. As a result, data validation, monitoring the model, cyber security controls and accountability to decision should be included in the CEB solution.

4.5 Ethical, governance and practical considerations

For CEB the peak demand analytics system will not only be a tool for technical forecasting. Because electricity is a national service that is critical. With ethical, governance and the practical controls the system must be also designed. If smart meter data, demand of customer segment, the weather feeds, the generation records or the risks on reliability is used in model. Privacy could be created by poor data handling and cyber security and risks on reliability. So, from the beginning the governance should be considered than addition after model development.

Consideration	CEB relevance	Control
Data privacy	Smart-meter or customer-segment data may reveal behaviour patterns.	Use aggregation, anonymization and access controls.
Cybersecurity	Energy-sector systems are critical infrastructure.	Separate analytics access from operational control systems and apply strong authentication.
Model drift	Demand patterns, tariffs, fuel prices and weather patterns may change.	Retrain and monitor the model regularly.
Explainability	Planners must understand why a risk alert is produced.	Show feature drivers, confidence and recent trends.
Human oversight	Incorrect automated decisions can affect reliability.	Keep the system as planner decision support, not automatic dispatch.
Data quality	Missing or incorrect readings can mislead forecasts.	Apply validation rules, outlier checks and audit logs.

Table 20 : Ethical, governance and practical considerations

It's shown in the table that simply the main risk isn't if the model is able to predict the peak demand risk. Bigger issue is that if CEB can use model responsibly and safely in a real operational setting. For instance, customer segment or smart meter data should be aggregated then anonymised so that no exposing of the individual behaviour of the individual. Because an analytics platform is connected to data in the energy sector cyber security is also critical it could become a weakness if it was not kept separate from the operational control systems. Another practical concern is the model drift. Because the behaviour of demand, the tariffs, the fuel prices and the weather patterns might change with time. Hence, regular monitoring, retraining and the audit logs should be included in the system.

Rather than as an automatic dispatch system the proposed solution should remain as a decision support system most importantly. For high risk conditions identification, a ML model can be used but fully understanding the availability of generators, the constraints in transmission, emergency failures or judgements in engineering cannot be done. For ensuring that the decision quality is improved by the analytics without creation of

dependence that is blind on predictions that are automated oversight of human planner is necessary.

Chapter 5 - Discussion, Interventions, Recommendations and Conclusion

5.1 Discussion of key findings

Overall finding is that before it's a modelling problem the CEB peak demand planning is a decision problem. Why this problem matters now is shown in the evidences in the Chapter 1. In the chapter 2 its shown that in to the demand, the peak pressure, the renewable variability, the thermal dependency and planning that is data driven the problem can be broken down. how do these concepts converted to analytical flow that is practical is demonstrated in Chapter 3. Once the smart meters, the SCADA feeds, the weather systems, the generation records and the demand patterns of customers are included, why the real implementation would be a big data problem is explained in the Chapter 4.

It's also shown in the report that for decision analytics that are good both organixational and technical thinking are needed. Classification of risks and identifying the important drivers can be done technically by the Random Forest model. While organizationally only if it's communicated through a dashboard the results become useful which connects to the responsibilities of planner and linked to the actions of intervention. This is why rather than ending with the accuracy of the model the discussion is naturally lead to recommendations. The goal is not building a model for the own sake of it. For CEB, improving the quality, the timing and the generation planning decision that is transparent is the goal.

In this case study its shown that the peak demand planning problem in CEB is suitable for the decision analytics since credible public evidence is there for it, variables that are measurable and decision consequences that are real. Also, in this report its shown that only by demand forecasting this problem can't be solved. From the interaction between the demand growth, the night peak pressure, the renewable variability, the conditions in hydro and dependency on thermal generation the planning risks are emerged.

The modelling logic is supported by the findings from EDA since around the official anchor value of 2024 the daily demand varies. For classification the risk classes are usable and a visible relationship is there between daily demand and night p\peak demand. Also, the case is supported by the ML result since 68.48% accuracy was received by the Random Forest Model and practical drivers like thermal generation, Hydro generation, daily demand, temperature and rainfall were identified.

Nevertheless, critical interpretation of results is needed. A major issue is the high risk recall issue because in an operational perspective missing the high risks days is costly.

This means that as a proof of concept the model is useful but not sufficient enough for deploying directly. Real CEB hourly data or half hour data, availability of generator, record

availability, weather marching that is better and the evaluation that is cost sensitive should be used in a production solution

5.2 Intervention strategy

Intervention	Purpose	Expected decision value
Peak-risk dashboard	Display Low, Medium and High-risk days with explanations.	Improves planner visibility before peak periods.
Weather-integrated forecasting	Combine temperature, rainfall and solar indicators with demand history.	Improves understanding of renewable and demand uncertainty.
High-risk alert protocol	Create operational review steps when High-risk days are predicted.	Reduces the chance of missing critical preparation.
Renewable balancing review	Use solar, hydro and thermal indicators to plan supply mix.	Supports cleaner and more reliable generation planning.
Data quality monitoring	Track missing values, delayed feeds and outliers.	Improves trust in the dashboard outputs.
Model performance monitoring	Review accuracy, recall and model drift over time.	Keeps the system useful as demand patterns change.

Table 21 : Recommended intervention strategy

5.3 Phased implementation roadmap

Phase	Action	Outcome
Phase 1: Evidence and data audit	Identify available demand, weather, generation and calendar data sources.	Clear understanding of data readiness and gaps.
Phase 2: Pilot dataset and dashboard	Build a pilot dashboard using historical verified data.	Demonstrates value without affecting live operations.
Phase 3: Model improvement	Test multiple algorithms and improve High-risk recall.	More reliable peak-risk classification.
Phase 4: Operational integration	Connect dashboard alerts to planner review procedures.	Analytics becomes part of daily decision making.
Phase 5: Governance and scaling	Introduce monitoring, cybersecurity controls and retraining.	Sustainable decision-support system.

Table 22 : Suggested implementation roadmap

Because CEB should not directly jump from an academic prototype to operational deployment this approach that is phased is important. The planners are allowed to test if the dashboard decisions improve, missing variables are identified and before scaling the trust is built by the pilot.

5.4 Recommendations

- With a pilot peak demand risk dashboard CEB should start. Which combine analytics that are descriptive, diagnostic and predictive.
- The risk class, the confidence level, the demand trend that us recent and the variables that are top contributing shown in the dashboard rather than only presenting the prediction of the model.

- In model evaluation the high risk recall should be given priority. Because, high risk day missing has operational consequences that are greater than the false alarms.
- Hourly or half hourly demand, dispatch, availability of generator, solar that is rooftop and weather data of CEB which are real should be used in future modeling when access is available.
- From the start into the solution the data governance should be built which includes the quality of data checks, access controls the monitoring of model and the audit logs.
- With the human planner oversight as a tool for decision support this system should remain. Because engineering judgement is required and, in the dataset, the operational constraints might not be there.
- To practical intervention actions like the demand response alerts, the maintenance timing, renewable balancing and backup generation review, CEB should connect analytics outputs.

5.5 Limitations

Main limitation is that practical dataset is synthetic and has been calibrated and its not internal operational data from CEB. About the limitations this report is transparent because SCADA which is confidential, smart meter and dispatch records are not available publicly. Model using data that is daily level is another limitation. While hourly half hour granularity is required for the real planning of electricity. Also detailed transmission constraints, the generator outages that are exact, conditions of fuel stock or regional demand categories are not included in this model.

The analysis is not being invalidated because of these limitations because demonstrating a decision analytics approach is the purpose. But operational strength of the results is limited by them however. The model shows how CEB could be supported by analytics, not that the model that is currently ready for live deployment for CEB is the honest conclusion.

5.6 Conclusion

For forecasting peak demand risk and generation planning improvement It's demonstrated in this report how the decision analytics can be used. Since it's a grounded in public evidence in Sri Lanka the case was selected. It's shown that in a changing and more complex planning environment the CEB is working by the total generation, the daily average generation, the max demand at night and the rooftop solar exports

The domain is first established in this report and before the aim is started the evidence and the problem context are there. The problem is broken down into keywords then (The 5W1H questions, the data factors, concepts of analytics, the 4V's and the considerations on feasibility).

A synthetic calibrated dataset was used for the practical solution for demonstrating a Random Forest peak risk classifier. 68.48% of accuracy was gained by the model and useful

feature importance insight, but high risk call that is weaker and the limitations of the system have to be looked after as a proof of concept that is academic.

The conclusion that is the strongest is that from a decision support system that is staged which combines the data integration, EDA, predictive modelling, explanations on dashboard, governance and intervention rule, the CEB would be benefited. Planners should not be placed by models. For seeing the risks earlier, understanding the drivers and better planning decisions that are evidence based it should be helpful.

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Appendices

Appendix A - Google Colab workflow summary

Step	Action	Purpose
1	Create synthetic calibrated daily dataset	Demonstrate CEB-focused analytics workflow without claiming confidential data.
2	Engineer demand, weather, generation and calendar features	Connect data columns to the CEB decision problem.
3	Create transparent peak-risk labels	Produce Low, Medium and High classes for classification.
4	Split the data using stratification	Ensure all risk classes appear in train and test sets.
5	Train Random Forest model	Learn non-linear relationships and produce feature importance.
6	Evaluate classification report and confusion matrix	Critically assess accuracy, recall and misclassification risk.
7	Interpret feature importance and recommendations	Translate model results into decision-support insights.

Table 23 : The Colab workflow summary

Appendix B - Source credibility matrix

Source	Why credible	How used in the report
CEB Statistical Digest 2024	Official utility publication.	Main evidence for generation, peak demand and rooftop solar figures.
CEB Long Term Generation Expansion Plan	Official planning document.	Supports the planning context.
PUCSL sources	Electricity regulator information.	Supports policy and renewable-energy context.
SLSEA sources	National sustainable-energy authority.	Supports renewable-energy context.
Reuters	Established international news source.	Supports reform and investment-pressure evidence.
scikit-learn documentation	Official library documentation.	Supports algorithm explanation.
Fayyad et al. (1996)	Academic KDD source.	Supports knowledge discovery discussion.

Table 24 : The Credibility of the main evidence sources